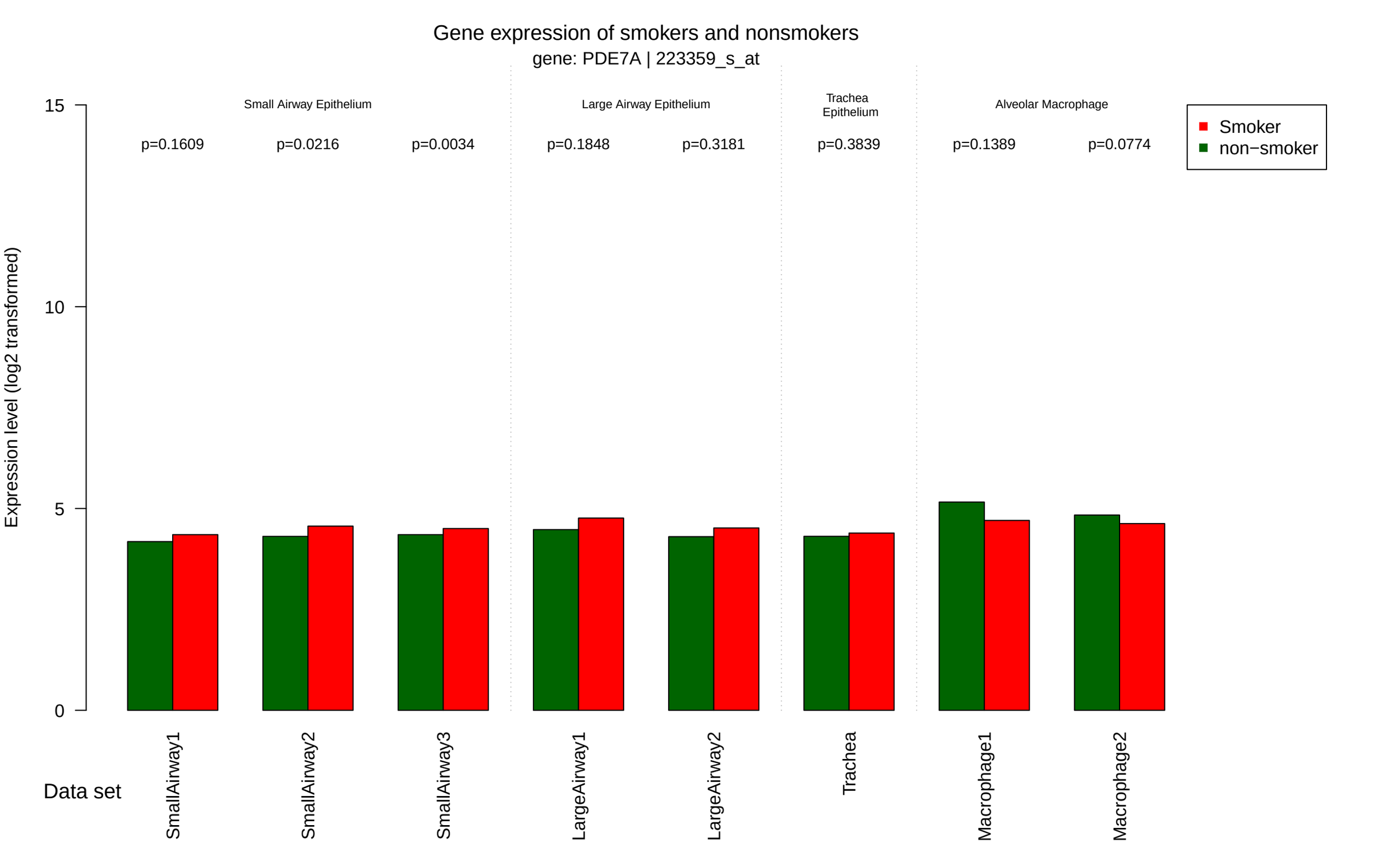
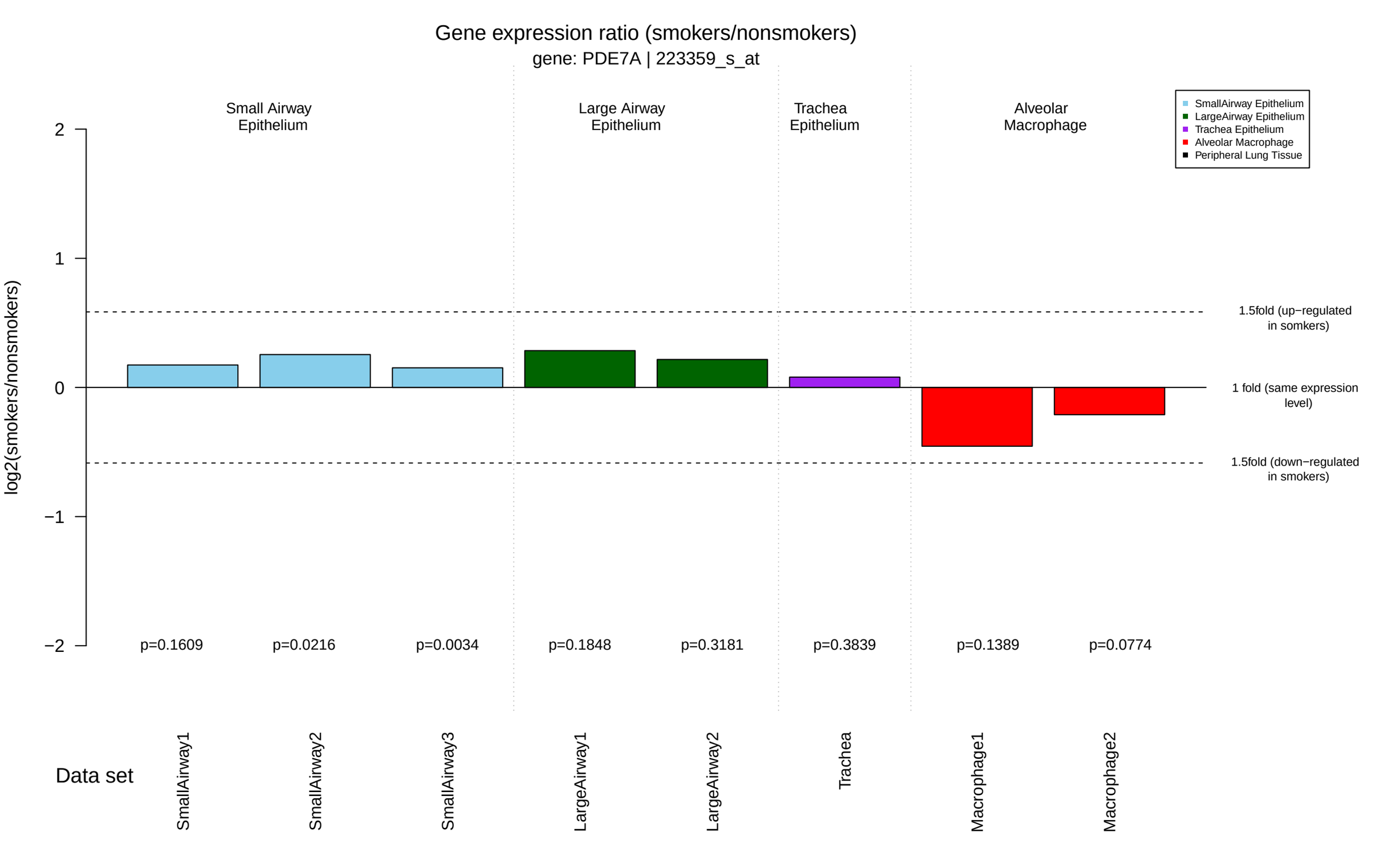
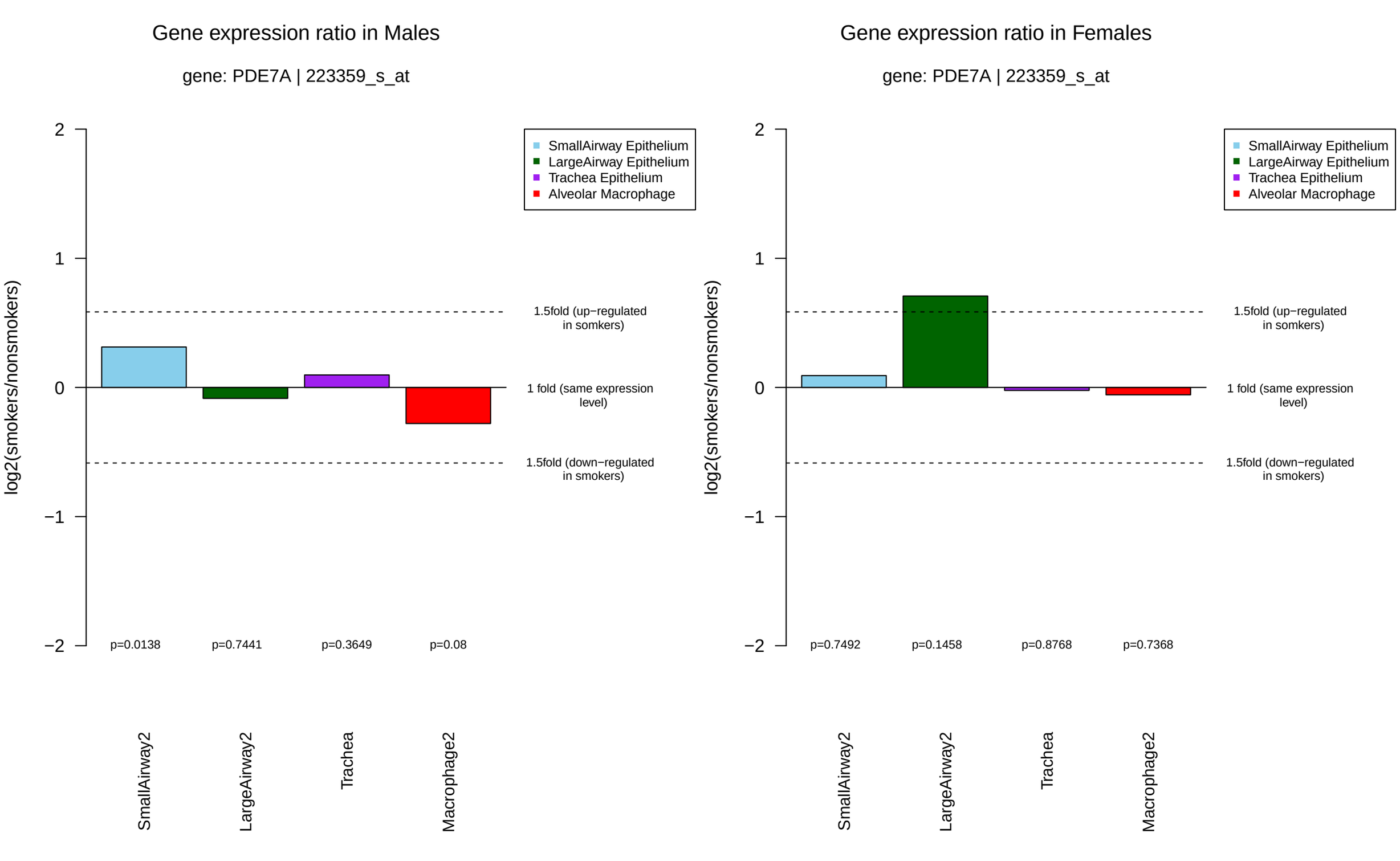
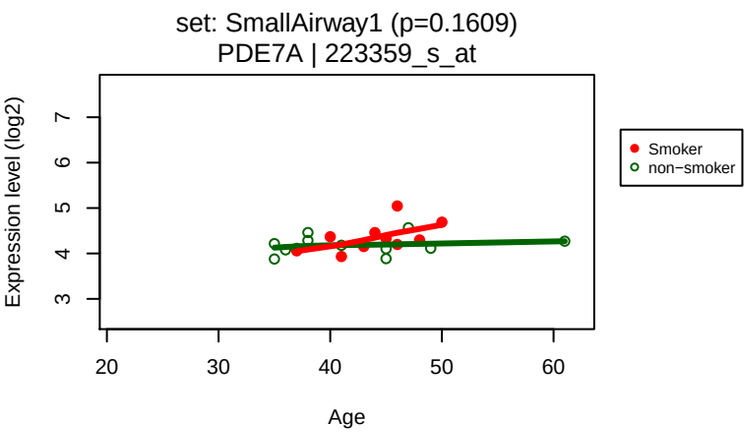
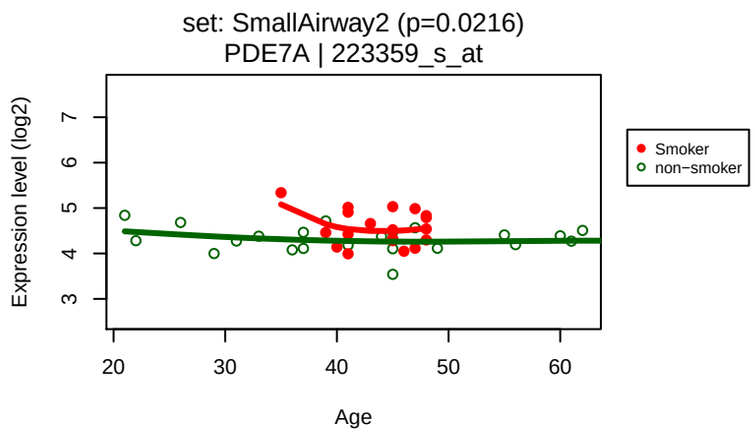
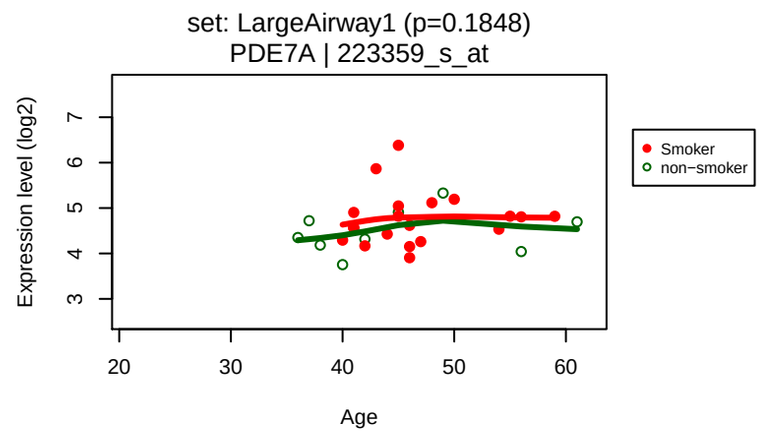

Gene expression of smokers and nonsmokers
gene: PDE7A | 223359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1609
p=0.0216
p=0.0034
p=0.1848
p=0.3181
p=0.3839
p=0.1389
p=0.0774
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

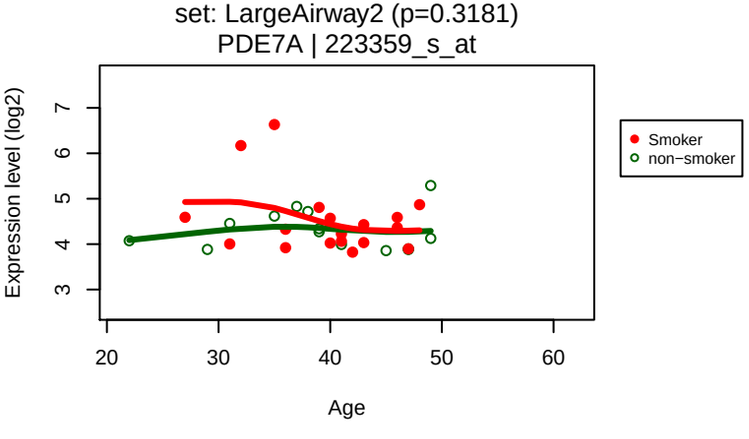

Gene expression ratio (smokers/nonsmokers)
gene: PDE7A | 223359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1609
p=0.0216
p=0.0034
p=0.1848
p=0.3181
p=0.3839
p=0.1389
p=0.0774
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

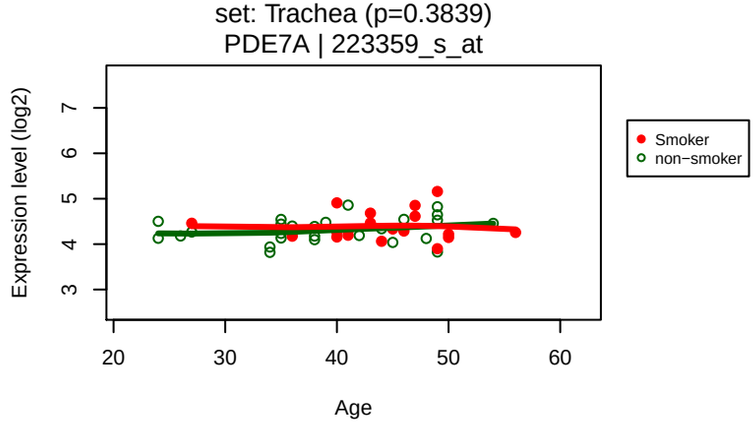

Gene expression ratio in Males
gene: PDE7A | 223359_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0138
p=0.7441
p=0.3649
p=0.08
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE7A | 223359_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7492
p=0.1458
p=0.8768
p=0.7368
SmallAirway2
LargeAirway2
Trachea
Macrophage2

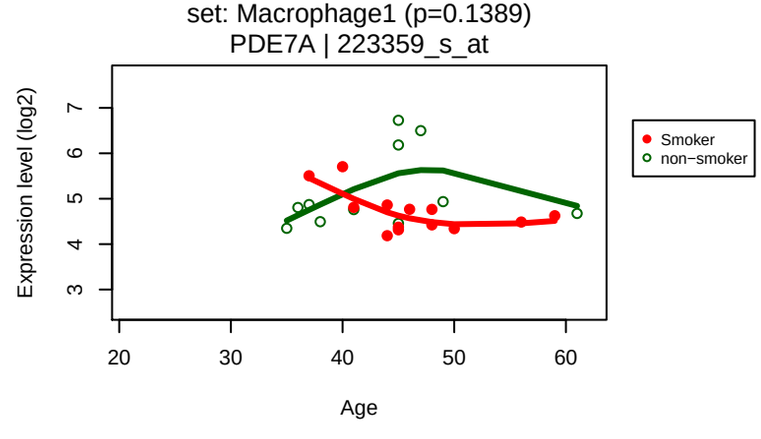

set: SmallAirway1 (p=0.1609)
PDE7A | 223359_s_at
Expression level (log2)
Age
Smoker
non−smoker

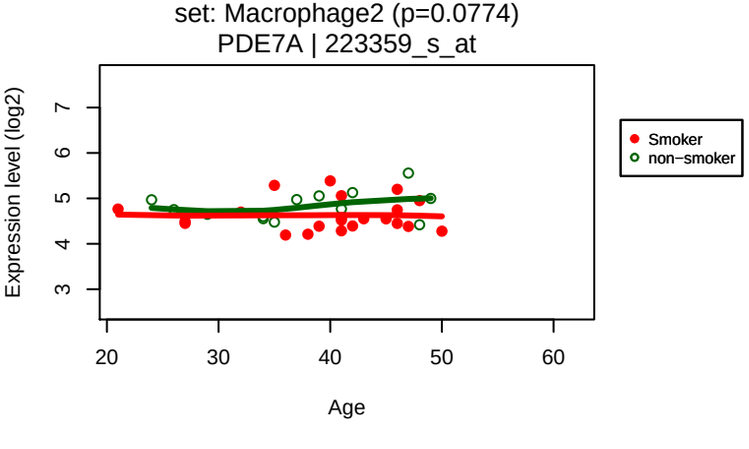

set: SmallAirway2 (p=0.0216)
PDE7A | 223359_s_at
Expression level (log2)
Age
Smoker
non−smoker

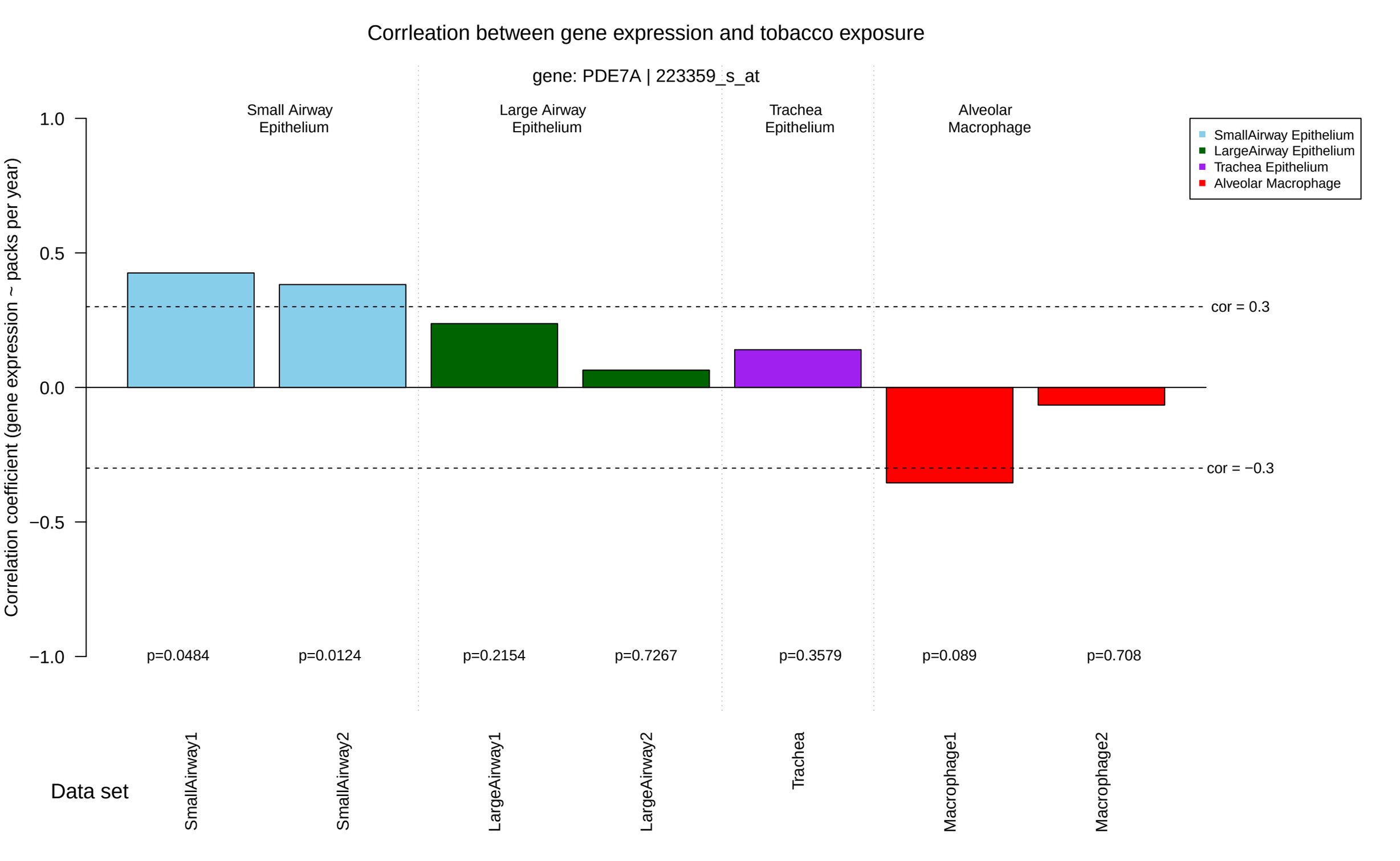

set: LargeAirway1 (p=0.1848)
PDE7A | 223359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3181)
PDE7A | 223359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3839)
PDE7A | 223359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1389)
PDE7A | 223359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0774)
PDE7A | 223359_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE7A | 223359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0484
p=0.0124
p=0.2154
p=0.7267
p=0.3579
p=0.089
p=0.708
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2